09F — Deep-Dive Feature Audit: Combat Engine & Tactical Simulation (Part 6 of 6 — Grand Finale)

Master Holistic Overview of the Entire Combat Ecosystem: `u.foe` Magic/Late-Game Stat Starvation, Uncapped `7`-Source Reflection Bypass, On-Map vs Off-Map (`DSH`) Split Universe, `O(N)` Hit-Loop Optimization, Mid-Fight Save/Load & Master `P0` $\to$ `P3` Roadmap

- **Target Codebase**: `JianghuSect 2` (`JS2-v2.6.0` in Google Drive `Js2 new`)
- **Target Audience**: Autonomous AI Developers (**GPT** & **Gemini**)
- **Role of Document `09F`**: **Master Holistic Synthesis** concluding the 6-part Combat Engine Deep Audit (`09A–09F`), connecting `js/combat.js`, `js/stats.js`, `js/skills.js`, `js/martial.js`, `js/cultivation.js`, `js/sim.js`, `js/explore.js`, and `js/expedition.js` into one unified, balanced, high-speed combat architecture.
- **Immutable Project Rules**:
    1. **NEVER modify `js/human_core.js`** (`HC.FINGERPRINT === '155:2f9c9dd7'`).
    2. **Preserve exact 40 `<script>` tags** and `<= 3.5 ms/tick` 80-NPC performance budget.
    3. **Zero RNG / Zero State Mutation During Pure UI & Preview Evaluation**.

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1. `[CMB-SYS-01]` Critical Enemy (`u.foe`) Stat Starvation & Buff Blackout (`js/stats.js:620–637` & `js/combat.js:840–900`): Why Magic Bosses & Tier 5–7 Raiders Collapse Against Geared Disciples!

Inspect `computeFoe` in `js/stats.js:627–636` verbatim:

```javascript
const P = u.power || 20, beast = u.kind === 'beast', t = u.tier || 1;
A.atk.b = P * (beast ? 0.62 : 0.55);
A.matk.b = P * (beast ? 0.25 : 0.2); // <-- ONLY 0.20*P (36% OF PHYSICAL ATK)!!
```

```
A.def.b = P * (beast ? 0.5 : 0.48) * (0.95 + 0.03 * t);
A.mdef.b = P * (beast ? 0.32 : 0.3); // <-- ONLY 0.30*P, ZERO TIER SCALING!!
...
A.pen.b = t; // <-- LITERALLY 1%..7% ARMOR PEN AT TIER 7 (VS 55%..82% ON
DISCIPLES)!!
```

Three Severe Holistic Balance Distortions:

1. **Enemy Caster / Demon / Spirit Bosses Hit Like Wet Noodles (`A.matk.b = 0.20 * P`)**:
    - Whenever an enemy sorcerer, ghost king, or elemental boss (`u.foe`) casts a `type: 'm'` technique, `C.strike` uses `sa.matk = 0.20 * P` against a disciple's heavily buffed `sd.mdef` (`~1.4 * P`)—dealing **`< 25%` of the damage of a basic physical bandit**!
    - **Fix**: Derive `matk` weight from the foe's archetype/element/skills (`caster/elemental/spirit foes: matk = P * 0.58, mdef = P * 0.48 * (0.95 + 0.04 * t)`; hybrid/boss foes: `atk = P * 0.56, matk = P * 0.54`)!
2. **Tier 5–7 Foe `pen` (`1..7`) & `mdef` Collapse Against Endgame Stacked Multipliers**:
    - Scale foe `pen.b = 3 * t + (u.isBoss ? 12 : 0)` (`up to 33% on Tier-7 Bosses`) and `mdef.b = P * 0.44 * (0.95 + 0.035 * t)` so late-game raiders and guardians remain genuinely dangerous!
3. **`computeFoe` Ignores `u.buffs` (`stats.js:635`) & `useSkill` Blocks `!a.foe` Buffs (`combat.js:840–900`)**:
    - `computeFoe` applies `statusMods(u, add)` (`debuffs!`) **while completely skipping `(u.buffs || []).forEach(...)`**, and `useSkill` blocks `ST.applyNamedBuff` with `!a.foe`! Allow Bosses/Elites to gain real named buffs (`tacticalCommand`, `astralMirage`, `karmicAegis`) when they cast support/movement skills!

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2. `[CMB-SYS-02]` Seven Unconsolidated Reflection Sources Bypass `C.reflectOf` (`35%` Cap), Exceeding `115%` Instant Reflect Per Hit (`js/combat.js:265–375, 1405–1435, 1505–1515`)!

While `C.reflectOf` (`lines 1405–1435`) cleanly caps `KW_REFLECT` + `karmicAegis` + `HEART_SOUL` + `KARMA_SAMSARA` + `TRAIT_PARRY_COUNTER` + `cryoMirror` at `35%` damage /

22% attacker hpMax, **seven other reflection hooks fire raw `C.applyDamage` calls outside `C.reflectOf`**:

1. BLACK_TORTOISE Guard Pet (22%, line 268)
2. VAJRA_IMMORTAL_BODY_BELL Mantra (16%, line 286)
3. SWORD_QI_MERIDIAN Mantra (12%, line 292)
4. PRIMORDIAL_YANG_FURNACE Mantra (8%, line 296)
5. DEFENSIVE_ARTS (TAIJI_SOFT_REDIRECTION / SWORD_CURTAIN / THORN_ARMOR, 18%–32% of **pre-mitigation `rawBefore`**, lines 321–331)
6. EFX_DAMAGE_REFLECT & EFX_V234_EMERGENCY_AEGIS (22% & 14%, lines 1506–1512)
7. LEG_WPN_打狗/PLONG (15%, visitors_clues.js:933)

- **Fix**: Feed passive percentage reflections (#2, #3, #4, #6) into C.reflectOf (allowing a soft diminishing cap up to 45% dmg / 22% Boss hpMax), gate melee counter-slashes (SWORD_QI_MERIDIAN, SWORD_CURTAIN, parryCounter, PLONG) on **`dist <= 2.4`**, and compute TAIJI_SOFT_REDIRECTION off mitigated dmg (capped by 0.18 * C.hpMax(a))!

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3. [CMB-SYS-03] Bridging the Split Universe Between On-Map Tick Combat (COMBAT.tick) & Off-Map Abstract Combat (evaluateTacticalBoutSummary in Sanctums / Pagoda / Sieges)!

Currently, JianghuSect 2 runs two half-disconnected combat worlds:

- **Systems Active ONLY On-Map (`0%` Effect in Sanctums, Pagoda, Sieges & Clue Fights)**: MANTRA_SYSTEM, DEFENSIVE_ARTS, SUPPORT_ARTS, FORBIDDEN_ARTS, PASSIVE_RESONANCE, DUET_CATALOG, PATH_GAUGES, KEYWORDS_CATALOG, and COMPOSABLE_ENGINE (34 2E/3E/4E Elemental Fusion Reactions + 8 Domains).
- **Systems Active ONLY Off-Map (`0%` Effect in Raids, Duels & Wild Bosses — Fixed in `09D`)**: evaluateFormationSynergy + CROSS_PATH_SYNERGIES.
- **Unified Bridge (`COMBAT.evaluatePartyCombatKitBonus(W, party)`)**:
  - Inside COMBAT.evaluateTacticalBoutSummary(partyA, partyB) (used by Sanctums, Pagoda, Sieges, and Rival Encounters), include a bounded 0.92..1.25× modifier reflecting:
    1) Bonded Duet pairs present in the party (+6%),
    2) Multi-elemental reaction coverage (>= 3 synergetic elements unlocking 3E/4E fusion +6%),
    3) Active Mantra / Defensive / Support / Constellation mastery (+0%..+8%).

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4. [CMB-SYS-04] Inner-Loop O(N_sect) Performance Hotspot in C.applyDamage (js/combat.js:308, 339, 349) & Mid-Fight Save/Load State Gaps (js/sim.js:560)!

1. **Eliminate `W.npcs` Full-Sect Scans Inside `C.applyDamage` (`combat.js:308, 339, 349`)**:
   - Every hit (and AoE/flurry sub-hit!) currently loops over **all `80–100` `W.npcs` up to 3 times** (`lines 308, 339, 349`) to look for nearby `DEFENSIVE_ARTS` guardians and `SUPPORT_ARTS` auras!
   - **Fix**: Iterate **only `C.alliesOf(W, f, unit.id)`** (`typically 1–6 active combatants in the fight`, falling back to `W.npcs` only if `!f`), reducing per-hit aura checks by **`15×–40×`**!
2. **Mid-Fight Save/Load State Preservation (`js/sim.js:560`)**:
   - Add `'_elemAuras'`, `'_poiseGauge'`, `'_ccDrStacks'`, `'_shadowClones'`, `'_boneArmorHp'`, and `'_supHealIcd'` to `PERSISTED_UNDERSCORE_KEYS` in `SIM.serialize` (`sim.js:560`) so saving and reloading mid-raid preserves active elemental reaction gauges, boss poise damage, ninja clones, and bone armor identically!

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5. Master Prioritized Execution Roadmap for GPT & Gemini (P0 $\to$ P3 Across 09A–09F)

Phase	Priority	Target Files	Exact Deliverables Across `09A–09F`
Phase 1	**P0 — Core Math Paradox, 15-Day Tribulation Clock & DoT Self-Harm Cascade**	`combat.js`, `stats.js`, `sim.js`	1) Anchor `C.strike` damage to stage-reference baseline HP (`refHp * frac`) instead of defender's buffed `C.hpMax(d)` so `+HP` genuinely increases hits-to-die without inflating incoming

Phase	Priority	Target Files	Exact Deliverables Across 09A–09F
			damage or crippling flat heals (09A). 2) Move CULT.TRIBULATION.tickAll(W) from SIM.dailyUpdate (15 days!) into SIM.tick (15 minutes!) (09E). 3) Tag C.statusTick DoT calls (note = 'DOT:' + x.k) so internal poison ticks stop draining the poisoner's Forbidden-Art HP, stop firing cross-map Taiji/Bell laser reflects, and preserve posthumous killer credit (09C).
Phase 2	**P1 — Spatial Physics, 24-Tile Row Block, Spar Leaks & ST Split-Brain**	combat.js, stats.js, skills.js	1) Unify ST.compute (explain) === ST.getBaseFast (get), fix _hpf denominator, apply it.combatMult/defMult, and wire TRAIT_FORMATION_STRATEGIST + TRAIT_VANGUARD_GUARD ally aura (09A). 2) Apply ST.get(u).move + zocSlowMult to

Phase	Priority	Target Files	Exact Deliverables Across `09A–09F`
			sect NPCs in `moveToward`, tighten `ROW_BODYBLOCK` from `24.0` $\to$ `3.0` tiles, fix early `return` on `evade`/`blink` so Archers actually step back and clear `_closedOnBy`, and block `BLACK_THREAD` sworn-enemy bodyguarding (`09B`). 3) Guard `KW_LAST_BREATH` (`+10 heartDemon`) & `TABOO_DESPERATION_NOVA` behind `f.serious`, cap cheat-death layers at `2`/fight, fix `BLOOD_HALT` ally-heal vs `healBlock` bug, and prevent jailing self-defense `grudge` victims (`09C/09E`).
Phase 3	**P2 — Foe Scaling, Reflect Consolidation, Summons & Raid Core-Breach AI**	`stats.js`, `combat.js`	1) Scale `computeFoe` `matk`, `mdef`, `pen` & `u.buffs` properly for caster/high-tier foes; consolidate the 7 external reflect hooks into `C.reflectOf` +

Phase	Priority	Target Files	Exact Deliverables Across 09A–09F
			melee distance gates (09F). 2) Clean delete W.foeById[sId] on summon cleanup, step summons inside COMBAT.tick, implement destroySummons on 4E-04, make AoEs shatter 1-HP _shadowClones, scale pet assists with max(atk, matk), and pass synDmgMult * masBase in resolveChannelTick (09C/09D). 3) Fix raid foeStep so raiders march on main_hall when no defender is within 16 tiles (09E).
Phase 4	**P3 — On-Map/Off-Map Synergy Bridge, Performance & Full Tribulation Autonomy**	combat.js, sim.js, cultivation.js, sect.js	1) Activate cached evaluateFormationSynergy + 43-path CROSS_PATH_SYNERGIES on-map (09D) and evaluatePartyCombatKitBonus off-map (09F). 2) Replace W.npcs loops in C.applyDamage

Phase	Priority	Target Files	Exact Deliverables Across `09A–09F`
			with `C.alliesOf`, persist mid-fight underscore keys (`_elemAuras`, `_shadowClones`, `_poiseGauge`), register `MAT_BOSS_*` items, and wire autonomous major-breakthrough `TRIBULATION` + Stage-9 `beginCelestialTribulation` + crisis `invokeAncestralPhantom` (`09C–09F`).